

Smart India Hackathon 2026 — Campus Evaluation Master Q&A;

Arvind Gavali College of Engineering, Satara • Project: BIS Sahayak (SIH26107)

Project: BIS Sahayak (Dual-Persona AI Assistant)	Problem Statement ID: SIH26107 (Software)
Live App: https://sih2026-bis-assistant.vercel.app	GitHub: github.com/PhoenixOP231/BIS_Sahayak_SIH

1. Novelty and Innovation (Max: 15 Marks)

Criteria: Originality of the idea; creative approach; clear differentiation from existing solutions.

■ ■ **30-Second Examiner Pitch:** "Generic search engines like Google only return 100-page unsearchable PDFs, while general AI like ChatGPT frequently hallucinates incorrect numbers for life-critical safety tolerances. BIS Sahayak innovates with (1) Dual-Persona RAG architecture tailored for consumers vs. engineers, (2) Real BIS Scheme-I license verification registry detecting counterfeit markings, and (3) Zero-hallucination clause inspection drawer with Hindi/English voice assist."

- **Dense-Lexical Hybrid Search:** Combines 768-dim normalized dense vectors with domain token weighting in <15ms.
- **Progressive Disclosure UI:** 2-sentence executive summary first, with expandable technical tables for deep exploration.
- **Offline Resilient Architecture:** In-memory fallback index ensures zero downtime even during DB/API outages.

2. Relevance to Problem Statement (Max: 10 Marks)

Criteria: Understanding of the problem; alignment of proposed solution with stated requirements (SIH26107).

■ ■ **30-Second Examiner Pitch:** "Problem Statement SIH26107 demands dual assistance for both everyday consumers and MSME industries regarding Indian Standards, QCOs, testing limits, and ISI marks. We provide 100% 1:1 alignment with Consumer Mode (plain safety guidance, fake mark alerts) and Industry Mode (proof stress, burst pressure, routine test tables) across 21 pre-indexed standards."

- **Broad Economic Scope:** 21 authentic standards covering Civil (Steel, Cement, Pipes), Electrical (Cables, Sockets, Lighting), Food & Safety (Drinking Water, Cookers, LPG, Helmets, Toys, Footwear).
- **Regulatory Grounding:** Fully aligned with Bureau of Indian Standards Act, 2016 and Ministry of Consumer Affairs QCO orders.

3. Technical Approach and Implementation (Max: 15 Marks)

Criteria: Appropriate technology, architecture or methodology; quality of implementation; technical depth.

■ ■ **30-Second Examiner Pitch:** "We implemented an enterprise-grade multi-tier architecture using Next.js 15 App Router, React 19, TypeScript, and Tailwind CSS v4. The AI core uses Google Gemini 2.5 Flash with low-temperature (0.3) context gating. Vectors are indexed in Neon PostgreSQL with pgvector (768-dim embeddings). Real-time speech synthesis runs via Web Speech API."

- **Pipeline:** Query -> 768-dim Embedding -> Hybrid Cosine Sim (<15ms) -> Persona Prompt -> Structured Response.
- **Quality Assurance:** 15 automated Vitest unit tests verifying RAG citations, Hindi responses, and license validation.

4. Feasibility and Practicality (Max: 15 Marks)

Criteria: Technical and operational feasibility; realistic resources, cost, timeline and deployment plan.

■ ■ **30-Second Examiner Pitch:** "Our prototype is 100% built, tested, and deployed live at production readiness. Serverless edge functions and Neon PostgreSQL scale to zero when idle, resulting in near-zero monthly operational cost. The modular pipeline can ingest all 20,000+ BIS standards and connect to the official e-BIS API effortlessly."

- **Zero Cost Scalability:** Pay-per-use cloud serverless edge model.
- **Implementation Roadmap:** Phase 1 (Core Prototype Live) -> Phase 2 (20,000+ Standards Batch Ingestion) -> Phase 3 (Government e-BIS Portal Webhooks).

5. Prototype / Proof of Concept (Max: 15 Marks)

Criteria: Working functionality; completeness, reliability and evidence that the core concept can operate.

■ ■ **30-Second Examiner Pitch:** "We invite examiners to test our live deployed prototype on their smartphones right now at <https://sih2026-bis-assistant.vercel.app>. Features include multi-turn conversational AI with Hindi audio playback, live CM/L license verifier with authentic vector ISI badge, and interactive standards catalog."

- **Live Verification:** Test real license CM/L-8400123 (Prestige) vs fake number 11111111 to see counterfeit warnings.
- **Production Reliability:** Zero compile errors, 100% route coverage on Vercel.

6. Impact and Scalability (Max: 10 Marks)

Criteria: Potential social, economic or institutional impact; reach, scalability and scope for adoption.

■ **30-Second Examiner Pitch:** "BIS Sahayak protects 1.4 Billion consumers from domestic hazards (exploding cookers, electrical fires, toxic toys). It saves 63 Million MSMEs millions in consultancy fees, boosting Make in India. It drives citizen awareness and active downloads for the official BIS Care Mobile App."

- **Social:** Reduces household fire accidents and infant toxicity hazards.
- **Economic:** Cuts factory compliance assessment cycles and audit rejections.

7. User Experience and Accessibility (Max: 5 Marks)

Criteria: Ease of use; inclusiveness; interface quality; suitability for intended users.

■ **30-Second Examiner Pitch:** "Designed for Indian accessibility with 100% bilingual English & Hindi support, Web Speech audio assist for non-literate citizens, mobile-first responsive layout, and 1-click quick test prompt pills."

- **Inclusive Design:** High contrast, accessible fonts, zero heavy app installation required.

8. Sustainability and Ethics (Max: 5 Marks)

Criteria: Long-term viability; responsible use; privacy, safety, environmental and ethical considerations.

■ **30-Second Examiner Pitch:** "Safety-first zero hallucination architecture guarantees no incorrect limits for life-critical goods. Zero personal data collected, strict API key isolation, and serverless energy efficiency."

- **Ethical AI:** Strict source gating with verbatim clause citations for manual verification.

9. Presentation and Response to Questions (Max: 5 Marks)

Criteria: Clarity of pitch and demonstration; evidence-based answers; effective time management.

■ **30-Second Examiner Pitch:** "All answers are grounded in official Indian Standards (IS numbers) and government gazettes. Clear live interactive demo ready on mobile and screen."

- **Key Defense:** RAG low-temp gating prevents hallucination; 7/8-digit DB check catches fake ISI marks.

10. Teamwork and Multidisciplinary Contribution (Max: 5 Marks)

Criteria: Role clarity, coordination, participation and integration of diverse skills.

■ **30-Second Examiner Pitch:** "Seamless collaboration across AI/Data Engineering (RAG, Gemini 2.5, pgvector), Full-Stack Development (Next.js 15, Vercel), UI/UX Accessibility (Tailwind v4, bilingual dictionaries), and Regulatory Standards Research (BIS Act 2016, 21 IS standards, Vitest QA)."

- **Coordinated Execution:** All code modularized, documented, and version-controlled on GitHub.